

Credit Risk & Data Quality Analytics Platform

AN END-TO-END DATA ANALYTICS, RISK & DATA QUALITY
SOLUTION

PYTHON | SQL | MACHINE LEARNING | POWER BI | EXCEL

Leisha Nampally
B.Tech Computer Science Engineering

Business Problem, Objectives & Dataset

Business Problem

Financial institutions need accurate and reliable credit data to assess risk and support data-driven decisions.

Credit data may contain:

- Missing or inconsistent values
- Data-quality issues
- Different default patterns across loan segments
- Potential anomalies

Dataset

32,581 loan records | 12 variables | Target: loan_status

Key attributes:

Income • Age • Loan Amount • Interest Rate • Loan Grade • Loan Intent • Loan Burden • Previous Default History

Goal: Transform raw credit data into trusted, actionable risk insights.

Project Objectives

- Clean and validate credit data
- Analyze loan-default patterns
- Perform SQL-based risk analysis
- Build ML models for default prediction
- Develop interactive Power BI dashboards
- Establish a reproducible analytics workflow

Data Preparation, Quality & Analytics Workflow

Data Preparation

- Inspected missing values and duplicates
- Standardized data types
- Validated numerical ranges
- Checked categorical consistency
- Validated target variable
- Identified potential outliers
- Created an analysis-ready dataset

Data Quality Framework

- Completeness — 25%
- Uniqueness — 20%
- Validity — 25%
- Consistency — 20%
- Data Type Quality — 10%

Overall Data Quality Score

100%

Analytics Workflow

Raw CSV → Python ETL → Data Quality Validation → Clean Dataset → SQL Analysis → Risk & ML → Power BI

Risk Analysis & Machine Learning

Credit Risk Analysis

- Calculated overall loan default rate
- Analyzed default rates by **loan grade**
- Compared risk across **loan purpose and home ownership**
- Examined **income segments and loan burden**
- Analyzed **previous default history**
- Investigated interest-rate and age-based patterns
- Flagged **potential anomalies** using extreme-value screening

Risk Focus

Loan burden + interest rate + previous default history + borrower characteristics

Machine Learning

Default Prediction Models

- **Logistic Regression** — baseline classification
- **Random Forest** — non-linear classification

Evaluation

- Precision
- Recall
- F1-score
- ROC-AUC
- Confusion Matrix

Explainability

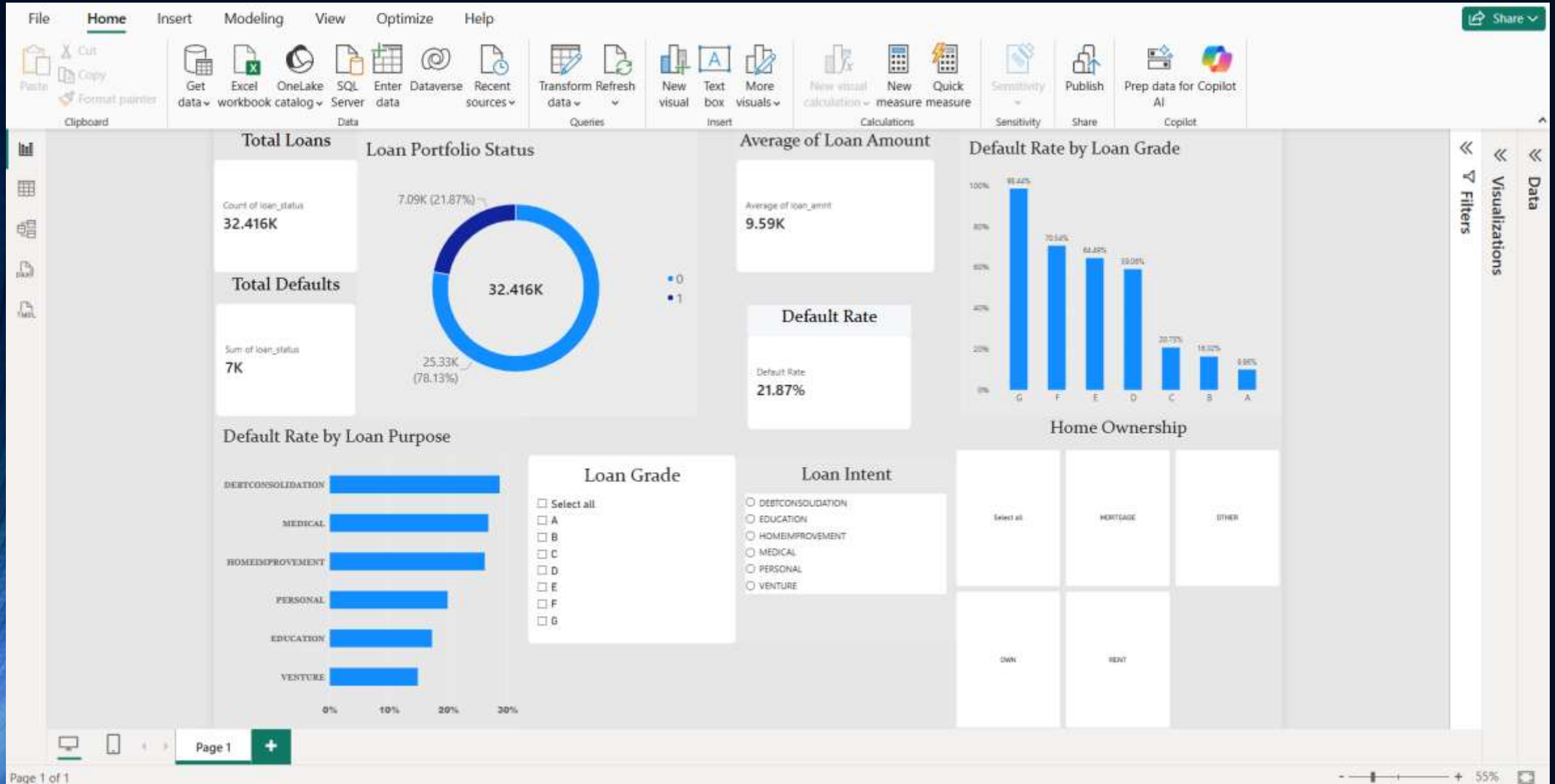
- Random Forest Feature Importance
- SHAP

Output:

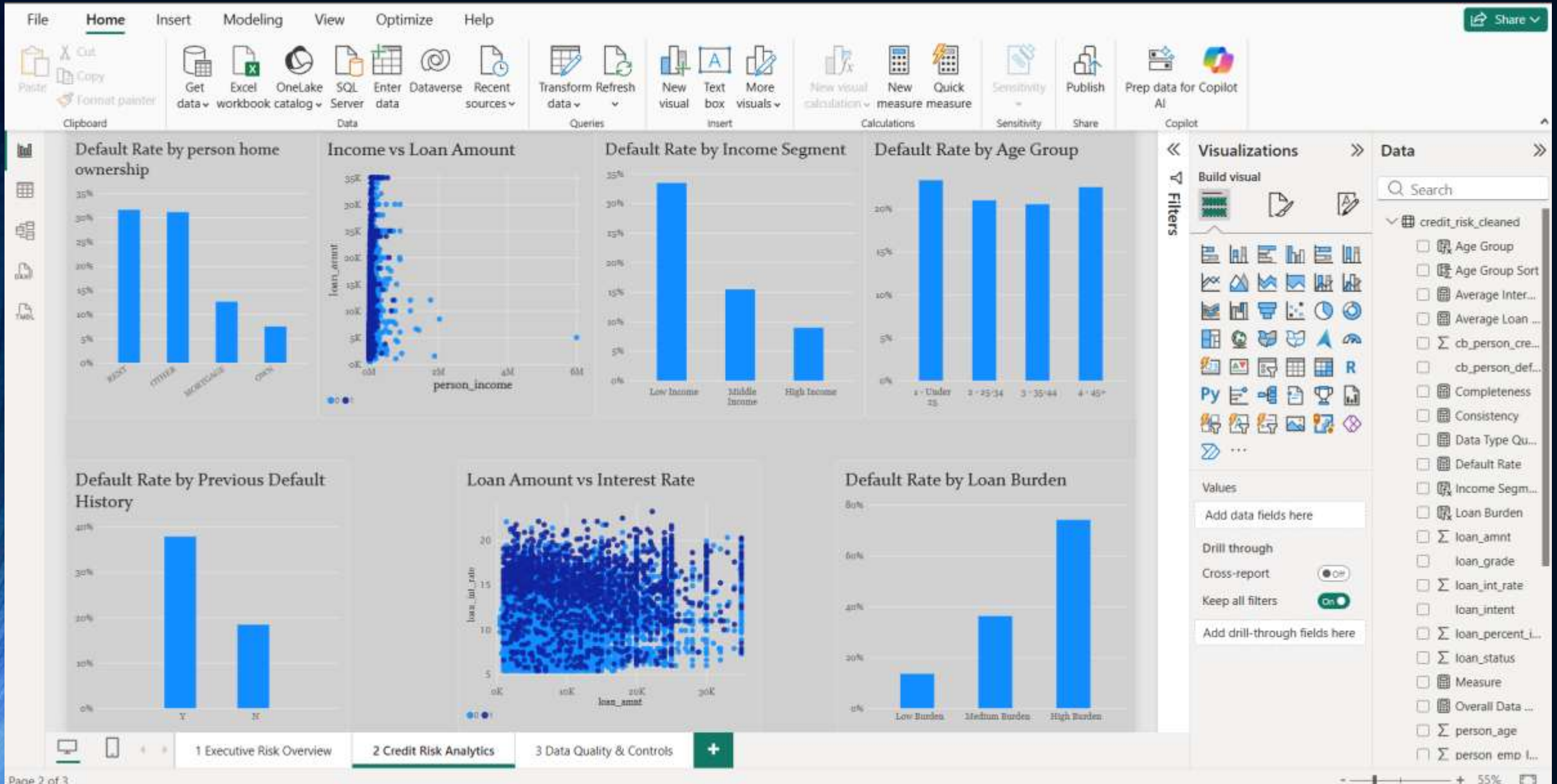
Predicted default probabilities and model-based risk indicators

EDA & Risk Analysis → Feature Preparation → ML Models → Model Evaluation → Explainability → Risk Insights

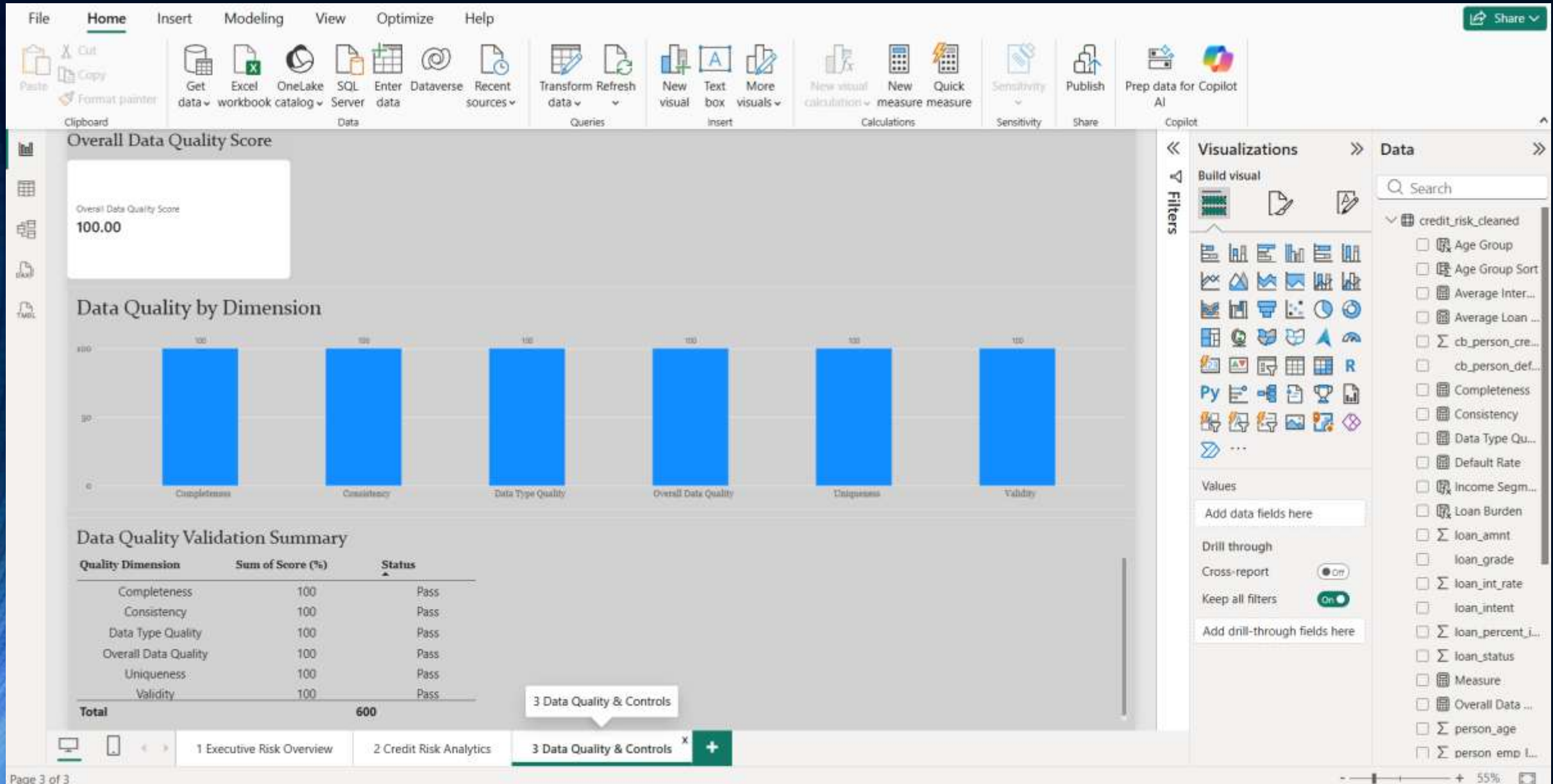
Power BI Dashboard



Power BI Dashboard



Power BI Dashboard



Key Insights & Project Outcomes

Key Insights

- Default rates vary across **loan grades and borrower segments**.
- **Loan burden** provides an important indicator for risk segmentation.
- **Previous default history** is a relevant risk signal in the analysis.
- Extreme-value screening identified **potential anomalies** for further investigation.
- Data-quality validation provides an additional layer of control before reporting.

Project Outcomes

- End-to-end **credit risk analytics workflow**
- Python-based data cleaning and EDA
- SQL-based analytical queries
- Data-quality assessment framework
- Logistic Regression and Random Forest models
- Model evaluation and explainability
- 3-page interactive Power BI dashboard
- Reproducible project structure with GitHub documentation

Conclusion

The project transforms raw credit data into validated analytical outputs and interactive risk insights by integrating Python, SQL, Machine Learning, Power BI and data-quality controls.